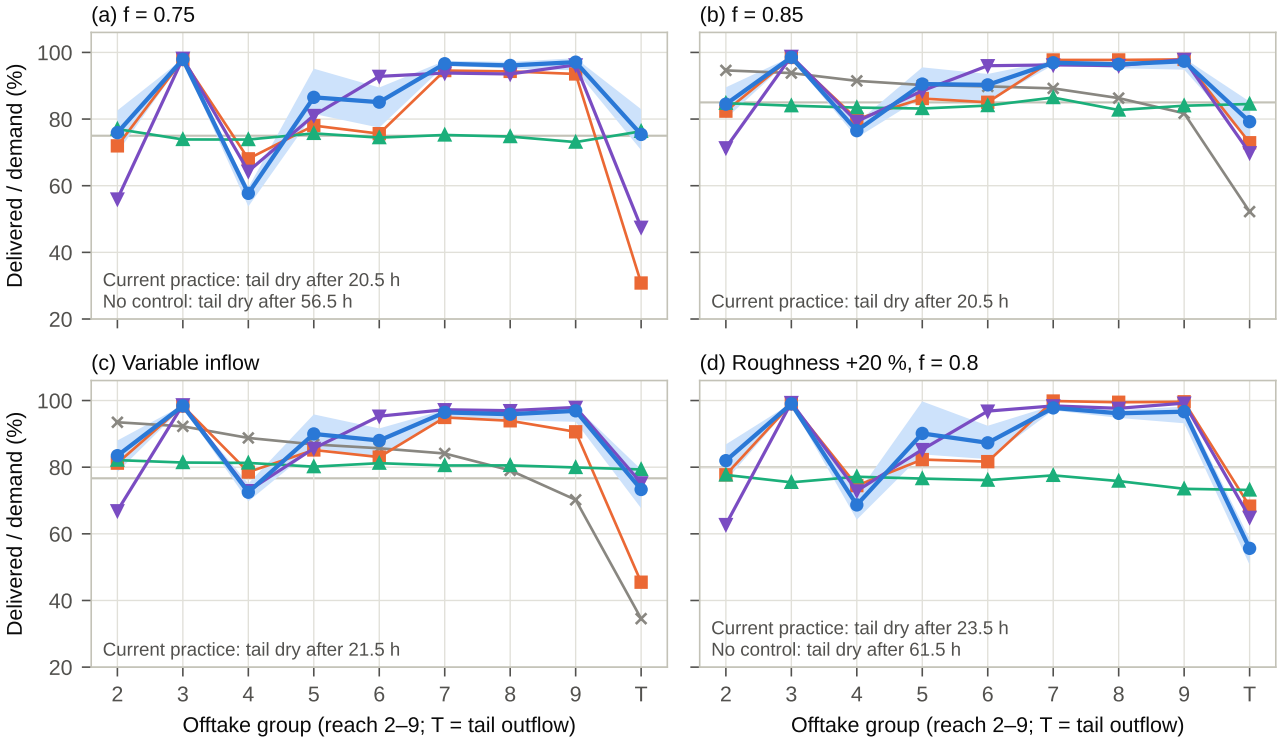
(a) f = 0.75
(b) f = 0.85
(c) Variable inflow
(d) Roughness +20 %, f = 0.8
Delivered / demand (%)
Current practice: tail dry after 20.5 h
No control: tail dry after 56.5 h
Current practice: tail dry after 20.5 h
Current practice: tail dry after 21.5 h
Current practice: tail dry after 23.5 h
No control: tail dry after 61.5 h
Offtake group (reach 2–9; T = tail outflow)
No control
Fixed rule + FF/P (13 gates)
Optimized staged rule + FF/P (13 gates), mean of 3 seeds
Neural + FF/P (13 gates), mean and range of 5 seeds
Full-control benchmark (66 gates)
Head supply ratio f